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Evaluation of the Maturity Level of the Process Safety Culture for Oil and Gas Industry Workers in a Region of Middle East

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Abstract

This paper evaluates the maturity level of the process safety culture within a major contracting organization in the oil and gas sector in a region of the Middle East, with a specific emphasis on leadership influence and human-centered perceptions. Using a questionnaire grounded in the Center for Chemical Process Safety (CCPS) core principles and the Hudson-Westrum maturity framework, responses from 172 workers—including managers, supervisors, and maintenance/construction staff—were analyzed to quantify perceptions of the existing safety culture. Results indicate an overall average maturity score of 74.5%, placing the organization in the "Calculative/Bureaucratic" stage. However, significant differences emerged between management and frontline workers, particularly in modules addressing leadership engagement and the prioritization of safety over production, underscoring the critical role of human-centered leadership in shaping these perceptions. The study found that while management often rated the process safety culture as proactive, supervisors and workers reported weaker implementation, particularly in areas such as visible leadership response to safety concerns and empowerment to prioritize safety. Conversely, open communication and trust mechanisms were reported favorably across all groups. The research underscores that achieving a proactive or generative safety culture requires bridging these perception gaps through consistent, demonstrated leadership commitment and strategies that embed process safety as a shared organizational value. Recommendations are provided for organizational leadership to implement unified safety vision strategies, enhance competency training, integrate safety metrics into performance reviews, and establish robust process safety frameworks to advance the culture maturity. These insights support practical leadership strategies to elevate the organization's process safety culture from calculative toward proactive maturity.

Introduction

Major incidents in the oil and gas industry continue to pose serious risks to workers, companies, and surrounding communities. Despite advances in engineering safeguards and regulatory frameworks, accidents still occur too often, especially in regions like the Middle East that rely heavily on oil and gas operations.

Data from the International Association of Oil & Gas Producers (IOGP, 2019a) (IOGP, 2023) highlights this challenge. They track Process Safety Events (PSEs) which are incidents involving unintended releases of hazardous materials or energy. These are classified by severity:

- **Tier 1** events are the most serious, often resulting in major consequences.
- **Tier 2** events are less severe but still reflect failures in safety barriers.

From 2014 to 2023, oil and gas operations in the Middle East reported between 1 and 40 Tier 1 process safety events annually, along with 12 to 83 Tier 2 events. This variation over nearly a decade underscores that the frequency of such incidents remains a significant concern for the industry as visible in *Figure 1*.

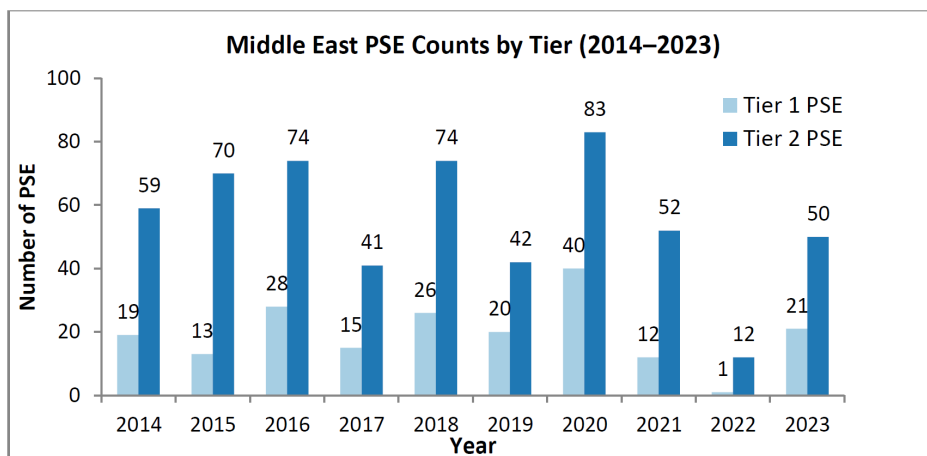


Figure 1—Reported Tier 1 and Tier 2 Process Safety Events in the Middle East oil and gas sector from 2014 to 2023 (adapted from (IOGP, 2019a) (IOGP, 2023)).

Investigations into past major incidents around the world, including offshore platform explosions and refinery fires, consistently reveal that most root causes are not technical equipment failures. Instead, they relate to communication and the everyday decisions that shape how work is performed, in short, to the process safety culture of organizations.

A review of global incident analyses resumed in *Figure 2* shows that organizational and cultural factors are the primary contributors in over half of major accidents.

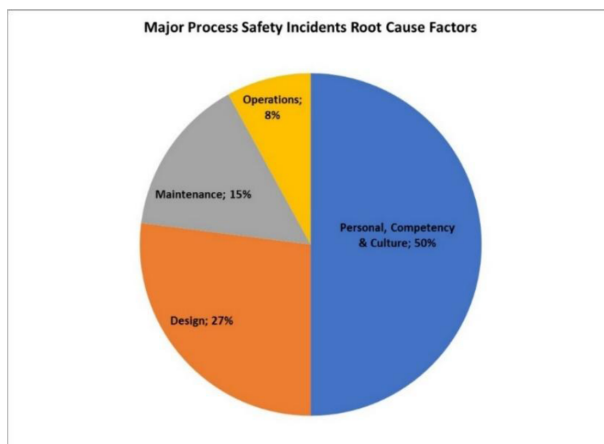


Figure 2—Root cause factors of major Process incidents from (ICHEME, (2020)).

This underscores a critical point: effective process safety is not achieved by technical measures alone. It requires a mature process safety culture in which leaders consistently prioritize safety over production, workers feel empowered to speak up and stop unsafe work, and risks are openly discussed and managed.

However, achieving this maturity is particularly challenging in the Middle East. Oil and gas projects here often involve diverse, multinational workforces, with differing cultural attitudes toward authority, communication, and risk. This can create gaps between how safety systems are intended to function and how they are actually implemented on site.

This study focuses on one of the largest contracting organizations operating in Oil & Gas sector across Saudi Arabia, Kuwait, and the UAE. It aims to evaluate the maturity of the company's process safety culture and understand differences in perceptions between managers, supervisors, and frontline workers.

Using a structured questionnaire based on recognized process safety culture principles, nearly 200 participants provided insights into leadership commitment, trust, communication, and how hazards are addressed. The findings indicate that while systems are in place and open discussion is generally supported, many employees still perceive that meeting production schedules often takes priority over strict process safety measures. Importantly, management tends to view the maturity of the process safety culture more positively than supervisors and workers, highlighting a disconnect that could undermine consistent implementation.

To address this, the study not only diagnoses current cultural gaps but also provides a structured roadmap for improvement. It applies the Westrum-Hudson process safety culture maturity model, a well-established framework in high-hazard industries (refer to 3.1.2).

By combining this model with a targeted questionnaire on core process safety culture principles, such as leadership commitment, mutual trust, hazard management, and communication, the study pinpoints specific areas that require advancement. It reveals where the organization currently stands on this maturity scale and identifies practical priorities, like strengthening leadership behaviors, empowering workers to intervene, and ensuring safety decisions consistently override production pressures.

Literature Review: Summary of Core Literature Findings

The concept of safety culture was first formalized following the Chernobyl disaster, emphasizing how organizational attitudes and shared beliefs influence safety outcomes (INSAG, 1991). Over time, this evolved into the more specialized notion of process safety culture, as defined by the Center for Chemical Process Safety (CCPS, 2007); (CCPSa, 2018), which integrates leadership commitment, trust, operational discipline, and continuous learning to prevent major hazard events.

Leadership has been identified as a central driver of cultural development. Schein's work on organizational culture underscores that leaders shape "the way we do things here," embedding priorities through both formal systems and informal signals (Schein, 1990) (NASA, 2003). To assess and improve cultural maturity, Hudson adapted Westrum's typology into a five-stage model from Pathological (where safety is ignored) to Generative (where safety is integral to operations), widely applied in the oil and gas sector (Hudson, 2007); (Filho & Waterson, 2018).

However, multiple studies reveal that management often perceives the maturity of process safety culture more positively than supervisors or frontline workers (Bordin, et al., 2013); (Alshahrani, et al., 2014). This disconnect can undermine trust and psychological safety, limiting open communication about hazards. It highlights why leadership must do more than communicate priorities superficially. As Schein (Schein, 2004) emphasized, "*the real thing of real importance that leaders do is to create and manage culture and that the unique talent of leaders is their ability to work with culture.*" By visibly and consistently prioritizing process safety over competing objectives, leaders can close these gaps, advance cultural maturity, and ultimately strengthen the well-being of their workforce.

Theoretical Framework and Definitions

The concept of process safety culture underpins this research. It is widely recognized that technical systems alone cannot prevent major incidents in high-hazard industries. The attitudes, beliefs, and behaviors that collectively define an organization's culture play a crucial role in managing the risks associated with hazardous materials and energy. This is particularly relevant in the oil and gas sector, where the complexity of operations and the diversity of the workforce increase the challenges of maintaining a consistently high standard of risk control.

Following the Chernobyl disaster, international bodies formally recognized that organizational culture could be a primary contributor to major accidents (INSAG, 1991). This led to the evolution of safety culture frameworks and, more specifically for this sector, to process safety culture models that integrate leadership commitment, hazard awareness, and open communication to prevent catastrophic losses (CCPS, 2007) (CCPSa, 2018).

Leadership's influence is foundational. Schein (Schein, 2004) emphasizes that culture is ultimately shaped by what leaders consistently prioritize, reward, or tolerate highlighting why leadership is a core focus in assessing process safety culture maturity.

Process Safety Culture

Core Definitions. **Process safety culture** refers to the shared values and behaviors that determine how process risks, those involving loss of containment of hazardous substances or energy, are understood and managed across an organization. Unlike personal or occupational safety, which focuses on individual accidents, process safety aims to prevent large-scale incidents that can result in multiple fatalities, significant asset loss, or environmental harm (CCPSa, 2018).

Maturity Models.

The Westrum-Hudson Model

To assess how deeply process safety is embedded in daily operations, this study uses the Westrum-Hudson maturity model. Originally developed by Westrum (Westrum, 1993) and adapted for the oil and gas sector by Hudson (Hudson, 2007), this model categorizes safety culture into five progressive stages:

1. **Pathological:** Safety is disregarded unless enforced.
2. **Reactive:** Action is taken only after incidents.
3. **Calculative or Bureaucratic:** Systems and metrics are in place, primarily for compliance.
4. **Proactive:** Hazards are actively sought out and managed before failures occur.
5. **Generative:** Safety is fully integrated into business operations, shaping all decisions.

This staged model allows organizations to diagnose where they currently stand and identify cultural attributes that must evolve to reduce the likelihood of major incidents (Filho & Waterson, 2018).

Assessment Methodology and Application of Tools

This research was conducted within one of the largest contracting organizations operating across Saudi Arabia, Kuwait, and the UAE. The study aimed to evaluate the maturity of process safety culture within this multinational environment, focusing on how leadership behaviors and organizational perceptions influence hazard management.

Survey Instrument and Structure

Questionnaire Development. The primary tool, employed in this study was a structured questionnaire explicitly designed to assess process safety culture. It was based on the ten core principles established by the Center for Chemical Process Safety (CCPSa, 2018), which include leadership commitment, mutual trust,

empowerment, maintaining a sense of vulnerability, and combating the normalization of deviance as shown in *Figure 3*.

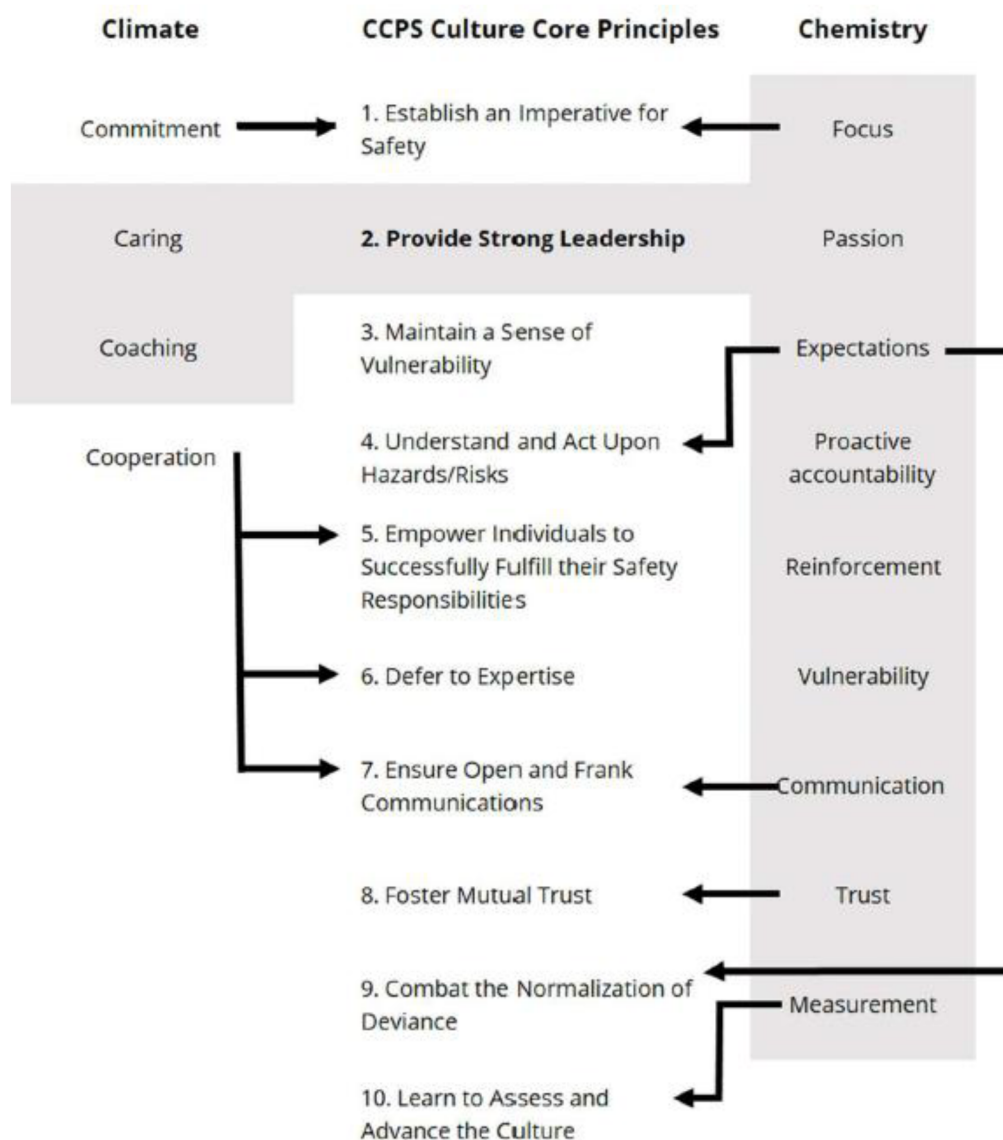


Figure 3—CCPS Process Safety Culture Core Principles (adapted from (CCPSa, 2018))

The questionnaire contained 26 items, carefully grouped into ten modules that directly reflect the ten core attributes of process safety culture as defined by the CCPS. Each module corresponds to a specific principle, such as leadership commitment, fostering trust, maintaining a sense of vulnerability, or combating the normalization of deviance. Within these modules, individual questions, referred to as factors, were crafted to probe perceptions and behaviors associated with each attribute. The total number of factors across all modules was 26. The ten modules are listed in *Figure 4*.

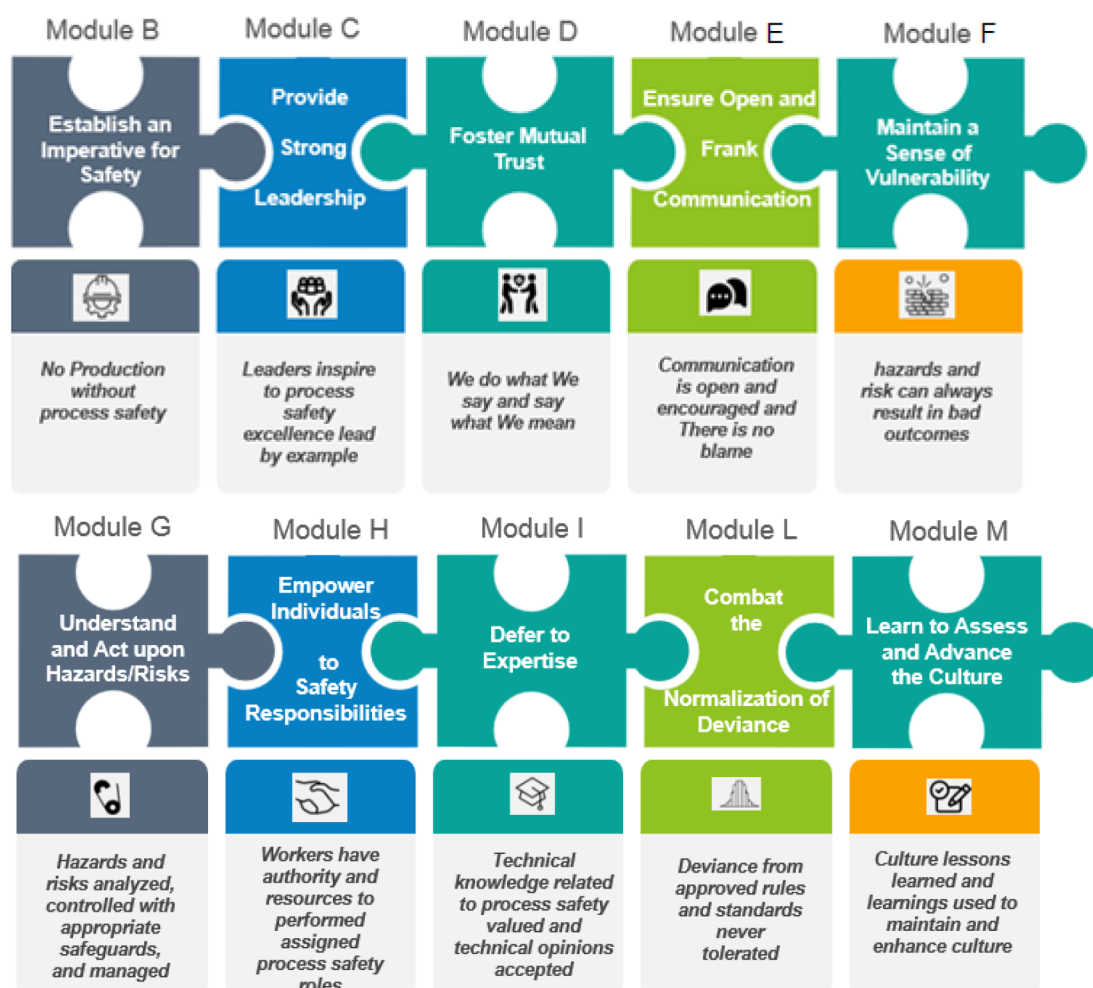


Figure 4—The ten Questionnaire Modules

Each item was assessed using a 5-point Likert scale, ranging from "Strongly Disagree" to "Strongly Agree," enabling a quantitative measurement of attitudes and perceptions related to process safety culture.

A pilot test was conducted with a local management consultancy to refine language for cultural neutrality and clarity, ensuring the instrument was contextually appropriate and understandable for a diverse workforce. This initial process helped strengthen the questionnaire's design and supported its subsequent validation, which is detailed in Section 4.2.1.1.

Application and Data Collection. The final survey was distributed to employees across management, supervisory, and frontline roles. In total, 172 valid responses were collected. Given the sensitive nature of discussing management priorities and potential conflicts, the survey was carefully worded to protect anonymity and encourage honest feedback.

Data Interpretation and Maturity Evaluation

Mapping to the Westrum-Hudson Model. The data collected through the structured questionnaire were systematically analyzed to evaluate the organization's process safety culture maturity. This analysis translated individual perceptions into meaningful indicators of how process safety principles are embedded across various organizational dimensions.

Statistical Acceptability of the Questionnaire Data

To ensure the robustness of the assessment, the questionnaire data were first tested for internal consistency. The reliability of the instrument was measured using Cronbach's Alpha, yielding an overall coefficient of

0.794, which indicates satisfactory internal consistency for studies of organizational culture (Bordin, et al., 2013). Module-specific alpha values generally exceeded 0.6, reinforcing the dependability of the individual scales. In addition, standard deviations were moderate (overall 0.32), reflecting reasonable agreement among participants despite their varied roles and backgrounds (see *Table 1*).

Table 1—Statistical summary of the process safety culture questionnaire, showing mean scores, standard deviations, internal consistency (Cronbach's α), and the number of factors (questions) for each module. This table summarizes the key attributes assessed by the study, reflecting the distribution and reliability of responses across all ten process safety culture modules and the total questionnaire.

Module	Mean	Stdev	Cronbach's α	Factors
Total Questionnaire	3.981	0.32	0.794	26
Establish an Imperative for Safety (Module B)	3.824	0.65	0.611	4
Provide Strong Leadership (Module C)	3.963	0.61	0.618	3
Foster Mutual Trust (Module D)	3.940	0.58	0.527	4
Ensure Open and Frank Communication (Module E)	4.171	0.53	0.612	2
Maintain a Sense of Vulnerability (Module F)	3.992	0.66	0.603	3
Understand and Act upon Hazards/Risks (Module G)	4.136	0.52	0.608	2
Empower Individuals to Safety Responsibilities (Module H)	3.944	0.76	0.604	2
Defer to Expertise (Module I)	4.125	0.58	0.668	2
Combat the Normalization of Deviance (Module L)	3.825	0.88	0.691	2
Learn to Assess and Advance the Culture (Module M)	4.087	0.70	0.665	2

By structuring the questionnaire around multiple factors (questions) for each module, the study further minimized the influence of any single response anomaly. As a result, the aggregated module scores provide a statistically reliable depiction of the organization's perceived process safety culture.

Applying the Maturity Model to the Data

The Westrum-Hudson maturity model, introduced in Section 3.1.2.1, served as the foundation for classifying the cultural maturity levels derived from the questionnaire. Once the mean scores for each of the ten modules were calculated, they were systematically mapped onto the thresholds defined by this model.

This process utilized an implementation and functionality matrix to categorize each module as pathological, reactive, calculative, proactive, or generative, based on established score ranges (*Table 2*). Through this approach, the study could determine the extent to which process safety principles are currently integrated into specific dimensions of the organization's culture, providing a clear and structured diagnostic without yet disclosing the actual maturity outcomes (which are detailed in the subsequent results chapter).

Each question of the questionnaire has been scored using the *equation 1* elaborated for 5 Points Likert Scale (*Murray, 2017*).

$$Score(\%) = \frac{Mean - 1}{0.04} \quad \text{Equation 1}$$

Equation 1 is based on the mean and score values represented in *Table 2*

Table 2—Conversion of Likert Scale Means to Scores

Response	Mean	Score
Strongly Agree	5	100
Agree	4	75
Neutral	3	50
Disagree	2	25
Strongly Disagree	1	1

The above *equation 1* suggests, for example, that a questionnaire's question obtains a mean of 3.8 represents a score of 70.

Once the scores for all individual questions (or factors) have been determined, the average maturity level of the ten attributes (or modules) of process safety culture can be assessed using the process safety culture maturity scale based on the Westrum-Hudson model (*Westrum, 1993*).

Table 3—Implementation and Functionality Table

Factor Score (Question score)	Process Safety Implementation and Effectiveness	Maturity Level
Score ≥ 85	The process safety factor exemplifies best practices and is fully integrated into routine operations	Generative
$75 < \text{Score} < 85$	The process safety factor is well established and reliably applied across activities	Proactive
$60 < \text{Score} \leq 75$	The process safety factor exists but is only partly effective in daily practice	Calculative
Score ≤ 60	The process safety factor is largely missing and does not effectively function	Reactive / Pathological

This matrix moves beyond numeric averages to explore how well process safety systems function in practice assessing whether cultural principles are only documented superficially or genuinely embedded in daily leadership, communication, and operational practices ensuring that the evaluation of process safety culture is not limited to quantitative means alone, but also considers practical effectiveness and depth of integration.

Use of the Relative Importance Index (RII)

In addition to maturity classification, the study calculated the Relative Importance Index (RII) for each factor (question) and aggregated these to the module level. This technique, adapted from the research of Jarkas and Bitar (Jarkas & Bitar, 2012), highlights the relative weight participants implicitly assign to different aspects of process safety culture.

The RII was determined using the following equation 2:

$$RII(\%) = \frac{5 \cdot (n_5) + 4 \cdot (n_4) + 3 \cdot (n_3) + 2 \cdot (n_2) + 1 \cdot (n_1)}{5 \cdot (n_5 + n_4 + n_3 + n_2 + n_1)} \cdot 100 \quad \text{Equation 2}$$

where n_1 to n_5 represent the number of respondents selecting each Likert option (from Strongly Disagree to Strongly Agree).

Higher RII values indicate that participants view a factor as more mature or robust within the organization, offering an additional perspective to prioritize areas for leadership attention and cultural enhancement.

This comprehensive methodology enables stakeholders to clearly recognize strengths, gaps, and areas for cultural enhancement. By integrating mean score conversion, rigorous reliability validation, structured maturity and implementation matrices, along with RII-based prioritization, it establishes a robust framework for assessing the current state of process safety culture. The specific insights and targeted improvement opportunities derived from this analysis will be presented in detail in the subsequent results chapter.

Presentation of Data and Results

This chapter presents the principal outcomes of the study, based on the structured assessment of process safety culture across the organization's being studied. Results are systematically organized to illustrate the maturity levels of key process safety culture attributes, differences in perceptions among hierarchical roles, and a consolidated evaluation of the organization's overall cultural positioning. Throughout, the findings are contextualized by reference to similar studies in the literature, reinforcing their relevance and credibility.

Process Safety Culture Maturity Across Attributes

Maturity Distribution by Module. The detailed maturity levels of the ten process safety culture modules are presented in *Figure 5*. Each module corresponds to a critical attribute drawn from the CCPS process safety culture framework, encompassing leadership commitment, mutual trust, maintaining a sense of vulnerability, hazard identification and response, empowerment, and learning.

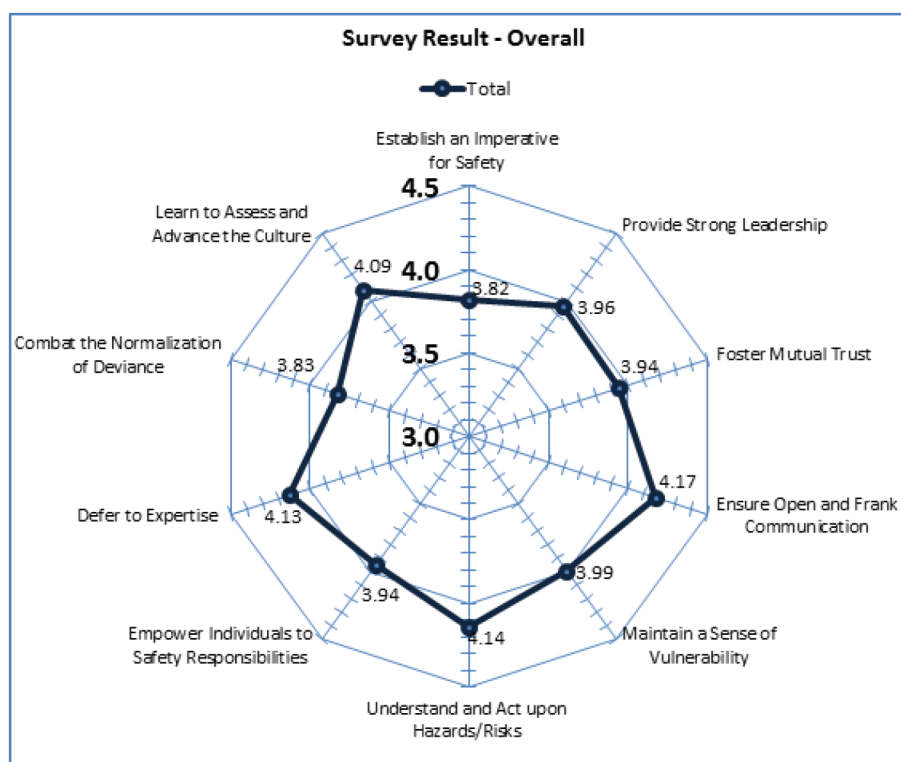


Figure 5—Survey Result - Overall by Mean (Min 1 – Max 5)

The distribution shown in *Figure 5* reveals that most modules are positioned within the calculative stage (mean between 3 and 4) of the Westrum-Hudson model. This indicates that formal systems and compliance mechanisms are generally well established, yet the cultural integration of process safety often remains procedural rather than intrinsic.

Notably, the modules "*Ensure Open and Frank Communication*" and "*Defer to Expertise*" achieved the highest mean scores, approaching the proactive boundary. This suggests that information sharing and reliance on technical input are relatively mature aspects of the organization's process safety culture. In contrast, "*Establish an Imperative for Safety*" and "*Combat the Normalization of Deviance*" displayed the lowest scores, indicating these areas may be more susceptible to being overshadowed by competing operational priorities.

Relative Ranking via Relative Importance Index (RII) Analysis. The comparative maturity of the process safety culture modules was further examined using the Relative Importance Index (RII), which ranks attributes by the relative weight ascribed by participants' responses. The outcomes of this analysis are presented in *Table 4*.

Table 4—Relative Importance Index (RII) ranking of process safety culture modules and factors.

Factor in summary (Refer Appendix A)	Score (%)	Implementation Level	Maturity Level	RII (%)
B1: Core Values awareness	77.33	Functional	Proactive	82
B2: Review of procedures before work	74.71	Partly functional	Calculative	80
B3: Correct analysis of changes	76.75	Functional	Proactive	81
B4: Safety over production	53.78	Not functional	Reactive	63
C1: Visible leadership	76.60	Functional	Proactive	81
C2: Accountability clarity	79.37	Functional	Proactive	83.5
C3: Reporting & solving problems	66.28	Partly functional	Calculative	73
D1: Report unsafe acts	79.22	Functional	Proactive	83
D2: Discuss safety before work	78	Functional	Proactive	82
D3: Challenge bad ideas	71.52	Partly functional	Calculative	77
D4: Use of safety metrics in appraisal	65	Partly functional	Calculative	72
E1: Speak up safety concerns	80.52	Functional	Proactive	84
E2: Access to safety info	78.05	Functional	Proactive	81
F1: Major risks analyzed	74.87	Partly functional	Calculative	80
F2: Safety is my responsibility	71.10	Partly functional	Calculative	77
F3: Manager monitors compliance	78.49	Functional	Proactive	83
G1: Implement JSA recommendations	74.71	Partly functional	Calculative	80
G2: Risks shared with contractors	82.12	Functional	Proactive	86
H1: Empowered to propose initiatives	74.72	Partly functional	Calculative	80
H2: Empowered to take corrective action	72.55	Partly functional	Calculative	78
I1: Rely on expertise during incidents	77.47	Functional	Proactive	82
I2: Rely on expertise before permits	78.78	Functional	Proactive	82.4
L1: Follow procedures w/o supervision	74.73	Partly functional	Calculative	80
L2: Never asked to operate unsafely	67	Partly functional	Calculative	73
M1: Learn from mistakes	76.47	Functional	Proactive	81
M2: Strategy to improve PSC	77.91	Functional	Proactive	82.3

The RII results closely mirror the mean maturity scores, highlighting communication and technical deference as relatively stronger attributes, while leadership imperatives and normalization of deviance are ranked lowest. This ordinal perspective is particularly valuable for guiding organizational priorities, allowing leadership to target specific areas for cultural enhancement. See Appendix A for the complete questionnaire results, which include detailed factor scores, module maturity levels, final rankings, and

overall cultural classification, providing a clear foundation for these targeted leadership and cultural enhancement priorities.

Comparative Perceptions by Organizational Role

Management, Supervisors, and Frontline Workers. *Figure 6* provides a breakdown of how process safety culture maturity is perceived across management, supervisory, and frontline levels.

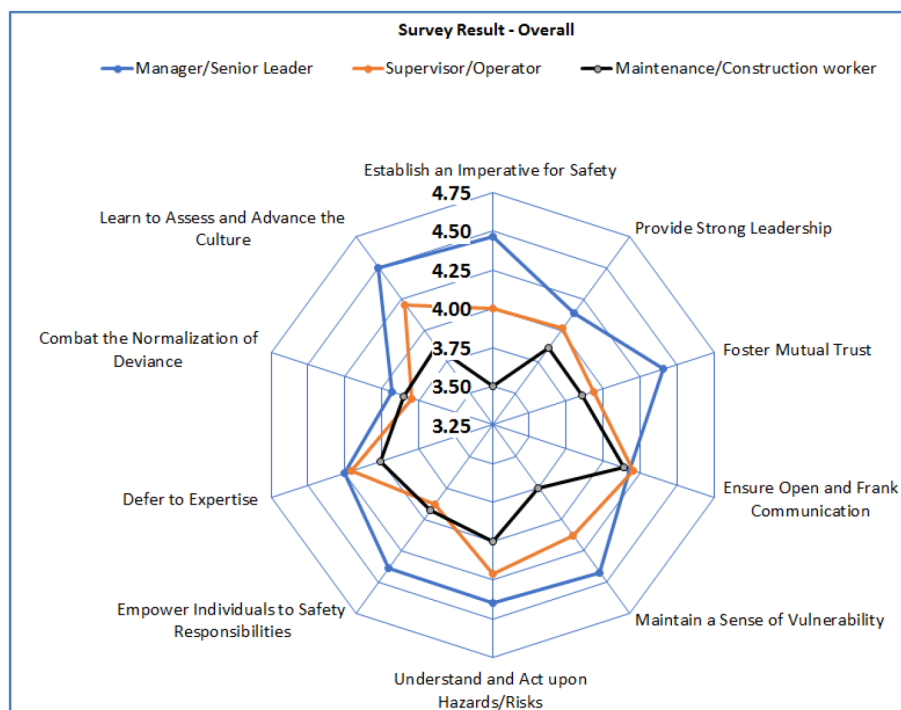


Figure 6—Survey Result - Overall by Role (Mean Value Min 1 – Max 5)

The data demonstrate a clear trend: management consistently perceives the process safety culture as more mature than supervisors and frontline employees. Supervisors occupy an intermediate position, recognizing the existence of systems but expressing reservations about consistent implementation. Frontline workers, closest to day-to-day operational risks, tend to hold the most cautious views.

This perception gradient is entirely consistent with prior studies. (Bordin, et al., 2013) and (Alshahrani, et al., 2014) documented similar divergences, noting that while management often considers safety culture robust, those directly executing work frequently observe gaps between procedures and practice. This disconnect can weaken mutual trust, discourage open reporting, and ultimately compromise psychological safety, all essential components of a truly generative process safety culture. These findings highlight the importance of addressing perception gaps to support psychological safety and cultural progression, with broader validation and parallels to existing research discussed in Section 5.4.

Production Pressures versus Process Safety Commitment. Another prominent insight from both the mean scores and RII data is that production demands continue to exert significant influence, at times perceived to surpass process safety considerations. This is particularly evident in factor B4 of the the module "*Establish an Imperative for Safety*" where lowest score suggest that operational objectives may occasionally overshadow rigorous process safety adherence.

This finding aligns closely with Schein's research (Schein, 2004), which clearly highlights that what leaders consistently emphasize and reward ultimately shapes organizational culture. When production milestones are perceived to receive greater managerial attention than process safety safeguards, the culture

is likely to plateau at a calculative level, constrained by compliance rather than driven by intrinsic commitment. This tendency often arises because managers regard safety primarily as a priority rather than a core value. Unlike values, which remain constant over time, priorities can shift in response to operational or commercial pressures, frequently to the benefit of production. In contrast, when process safety is genuinely embraced as a foundational value, it is insulated from such temporal shifts, ensuring that its importance does not waver even under intense production demands. Broader evidence and parallels from the literature reinforcing these dynamics are presented in Section 5.4.

Overall Process Safety Culture Positioning

The consolidated evaluation of the organization's process safety culture maturity is depicted in *Figure 7*, which plots the average maturity level derived from all modules and factors against the Westrum-Hudson thresholds.

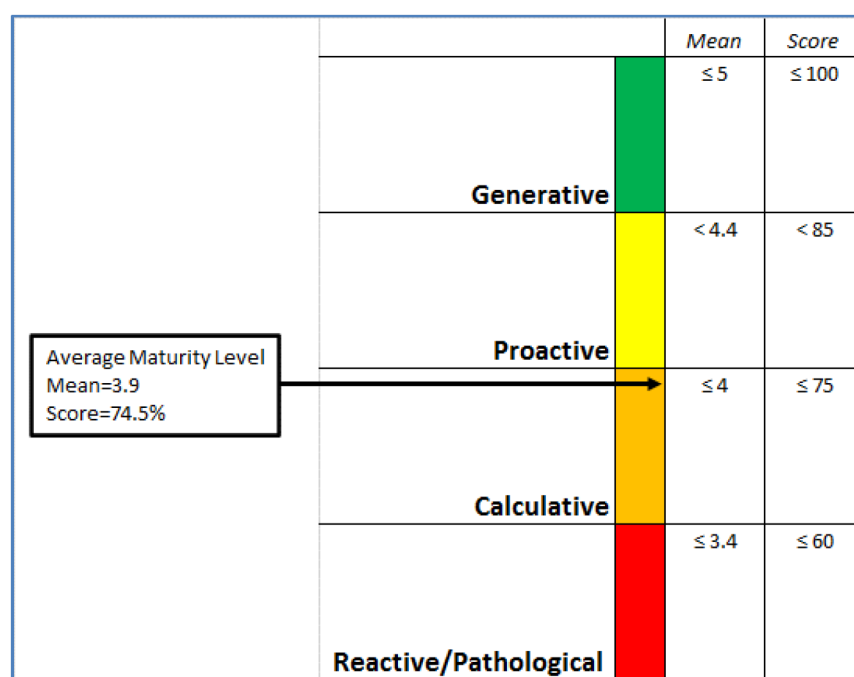


Figure 7—Maturity Level

This figure clearly situates the organization at the upper end of the Calculative stage, bordering the proactive threshold. With a mean of 3.9 and an equivalent maturity score of 74.5%, the data indicate that while there is a solid foundation of systems and formal processes, the culture has not yet fully transitioned into a proactive stance where process safety is instinctively prioritized across all decision-making contexts.

Alignment with Existing Literature

The results of this research reinforce prevailing insights in process safety literature.

The research's outcome is entirely aligned with global patterns reported by Hudson (Hudson, (2007)) and Filho & Waterson (Filho & Waterson, (2018)), who noted that many organizations stabilize at calculative maturity unless there is deliberate, sustained leadership intervention to embed process safety as a core organizational value.

The assessed maturity level of process safety culture concurs with other similar investigations conducted by Behari on gas plants in the Middle East (Behari, (2019)), Bordin in Oil and Gas Plants in Thailand (Bordin, et al., 2013) and Alasamri in Construction sites of Kingdom of Saudi Arabia (Alasamri, et al., 2012).

However, the author highlights that only a few numbers of published literature are reported on the development and application of maturity models for process safety culture (Shehata, et al., 2019) hence few data are available to benchmark the maturity of organizations process safety culture in the Middle East.

The observation that communication and technical expertise are relatively stronger cultural attributes is consistent with Forest (Forest, 2012), who found that these elements often mature earlier in multinational oil and gas operations due to the inherent necessity of procedural coordination.

Conversely, the identified challenges in sustaining a sense of vulnerability, resisting the normalization of unsafe practices, and consistently prioritizing process safety over production pressures reflect patterns emphasized by CCPS (CCPS, 2007), (CCPSa, 2018) and Hudson (Hudson, 2007). These studies underscore that advancing from a calculative to a proactive or generative culture demands far more than procedural compliance; it necessitates a visible and authentic leadership commitment that actively shapes what is discussed, rewarded, and systematically reinforced.

Although process safety culture is broadly regarded as an important element of daily operations, perceptions differ notably between groups, a phenomenon emphasized by Sharon (Sharon, 1999) and Bordin researches (Bordin, et al., 2013). The *Center for Chemical Process Safety* (CCPS) similarly acknowledges that "*process safety culture may not be completely uniform across a given company or site*" (CCPSa, 2018). Interestingly, local workers demonstrated a positive stance toward process safety culture in this study, as they did not score significantly lower than the overall averages. This finding aligns with Saad's study (Saad, 2016), which documented higher levels of safety behavior among local workers within the same regional context investigated by the author.

Consolidated Summary of Key Findings

Drawing these analyses together, the study yields the following principal conclusions regarding the state of process safety culture within the organization:

- The overall process safety culture resides at the high end of the calculative stage, characterized by well-developed systems but often driven by compliance rather than by intrinsic cultural integration.
- Attributes related to open communication and deference to technical expertise are relative cultural strengths, suggesting existing mechanisms that can be leveraged to advance maturity.
- Leadership imperatives for process safety, maintaining a sustained sense of vulnerability, and actively resisting normalization of deviance emerged as the least mature dimensions, pinpointing critical areas for organizational focus.
- There is a pronounced perception gap across hierarchical roles, with management consistently viewing the culture more positively than supervisors and frontline personnel. This highlights the need for leadership to more visibly and consistently engage across all levels, fostering trust and reinforcing that process safety remains non-negotiable even under commercial pressures.
- These findings align closely with established process safety culture studies, reinforcing the conclusion that leadership behaviors and the clear establishment of process safety as a core value, rather than just a shifting priority, are the pivotal levers to move beyond a calculative stage and cultivate a truly proactive or generative culture.

Conclusions

This chapter distills the essential conclusions of this research, drawn directly from the data and analysis presented in Chapter 5. These conclusions encapsulate the current state of process safety culture within the organization studied, spanning its operations in Saudi Arabia, Kuwait, and the UAE. They also emphasize the practical significance of these findings for both local and broader industry practice.

Process Safety Culture Maturity: Solid Foundations Yet Calculative Orientation

The research has conclusively established that the organization's process safety culture resides predominantly within the upper calculative stage, bordering on the proactive level of the Westrum-Hudson model (*Figure 7*). The consolidated maturity level, indicated by a mean score of 3.9 and an equivalent 74.5%, reflects a well-institutionalized framework of systems, compliance protocols, and formal procedures. For complete transparency and to support future benchmarking efforts, a detailed table listing all assessed factors, their individual scores, corresponding process safety culture modules, calculated maturity levels, and relative rankings has been included in **Appendix A**.

This demonstrates that the organization has invested significantly in developing structural elements of process safety, with documented processes, training programs, and monitoring systems forming an established baseline. However, as demonstrated by the scores across multiple modules (*Figure 5, Table 4*), this orientation is largely *compliance-driven*, with process safety often perceived as a regulatory or contractual requirement rather than an intrinsic cultural imperative.

Relative Strengths in Communication and Technical Expertise

Among the ten assessed modules of process safety culture, the attributes "*Ensure Open and Frank Communication*" and "*Defer to Expertise*" consistently achieved the highest mean scores, as well as the strongest RII rankings (*Table 4*). These modules approached the proactive threshold, indicating that mechanisms for transparent dialogue and reliance on technical knowledge are relatively robust within the organization.

Such strengths were particularly evident in factors like E1 (related to "Speak up safety concerns", 80.52% score, 84% RII) and G2 ("Risks shared with contractors", 82.12% score, 86% RII). These findings support Forest (*Forest, 2012*), who emphasized that communication and technical collaboration often evolve earlier in multinational oil and gas operations due to operational complexities requiring rigorous coordination.

This suggests that the organization possesses existing cultural assets that can serve as catalysts for broader process safety cultural advancement, provided these strengths are actively leveraged.

Critical Weaknesses: Leadership Imperatives, Normalization of Deviance, and Production Pressures

Conversely, the analysis highlighted significant deficiencies in modules such as "*Establish an Imperative for Safety*" and "*Combat the Normalization of Deviance*". Particularly telling was Factor B4 ("*Safety over production*"), which recorded the lowest individual score at 53.78% and an RII of 63%, situating it within the reactive stage.

This underscores a key vulnerability: while systems exist, there is a prevailing perception that production targets frequently override process safety considerations, especially under schedule pressures. This tendency often emerges because safety is treated as a priority, which by nature can fluctuate, rather than as a core value, which remains constant over time.

According to Schein (*Schein, 2004*), organizational culture is fundamentally shaped by what leaders consistently emphasize and reward. Without visible, unequivocal leadership emphasis on process safety as non-negotiable, the culture is likely to plateau, constrained by compliance rather than propelled by shared intrinsic commitment.

Pronounced Perception Gaps Across Organizational Levels

The study also revealed substantial differences in how process safety culture is perceived across hierarchical roles (*Figure 6*). Management consistently viewed the organization's culture as more mature, while supervisors registered more cautious assessments, and frontline workers, those most intimately exposed to operational hazards, reported the lowest confidence levels across many factors.

This gradient is not merely a theoretical observation; it is quantitatively supported by clear differentials in mean scores across roles for nearly all modules and highlight a critical cultural challenge: perception divides can erode mutual trust, dampen psychological safety, and inhibit the open communication essential to a proactive process safety environment.

Influence of Workforce Demographics: Local Worker Engagement

A noteworthy and constructive insight is that local workers did not score significantly lower than the overall average across the process safety culture attributes. This finding, discussed in detail alongside Saad's work (Saad, 2016), suggests that local employees within the studied regional context demonstrate a relatively strong alignment with process safety values and behaviors.

This counters some generalized assumptions in multinational settings that local workforces may require disproportionately intensive cultural reinforcement. Instead, it positions local engagement as a potential asset to be strategically cultivated through leadership visibility and participatory safety programs.

Alignment with Broader Industry Insights

Collectively, these findings are entirely consistent with global patterns as documented in 5.4.

Practical Utility of These Conclusions

The conclusions drawn from this research carry substantial practical implications for the organization under study and for similar high-hazard enterprises seeking to advance their process safety culture maturity.

1. **Targeted leadership action is imperative:** Without clear, consistent reinforcement that process safety is a core organizational value, not merely a priority, the culture is unlikely to progress beyond calculative maturity.
2. **Leveraging communication and expertise strengths:** These relative cultural assets should be deliberately harnessed to drive more challenging attributes, such as sustaining vulnerability and combating unsafe normalization, toward higher maturity.
3. **Closing perceptual divides:** Management must engage authentically across all organizational levels, ensuring that safety messages are not only delivered but visibly enacted, thereby building trust and enhancing psychological safety.
4. **Strengthening connection with the workforce:** To achieve a truly aligned process safety culture, management should remain closely connected with supervisors and frontline personnel, actively seeking to understand and reconcile any perceptual differences regarding the organization's cultural maturity.
5. **Integrating local workforce strengths:** Given the demonstrated positive attitudes among local workers, tailored participatory initiatives could further solidify process safety as a shared, non-negotiable value across the regional context.
6. **Benchmarking and continuous assessment:** This structured maturity analysis provides a replicable framework for periodic re-assessment, enabling the organization to track progress, adjust strategies, and systematically advance toward a proactive or generative process safety culture.

Concluding Remark

In sum, this study provides a rigorously data-backed evaluation of the organization's process safety culture, revealing solid systemic foundations tempered by calculative cultural tendencies, pronounced leadership and production-driven vulnerabilities, and clear opportunities for strategic cultural evolution. By translating these conclusions into focused leadership initiatives and reinforcing process safety as a core, immutable value, the organization is well positioned to elevate its culture beyond compliance fostering an environment where process safety is instinctively integral to every decision and action.

Study Limitations and Recommendations for Future Research

It has been demonstrated that there may be inherent limitations related to interpreting data derived from questionnaire-based safety culture assessments (Guldenmund, 2010). Although this study incorporated a pilot-tested and carefully reviewed questionnaire, there remains the possibility that participants may have interpreted certain items differently, potentially introducing unintended biases into their responses. Nonetheless, the author is confident that the iterative design process, including targeted feedback from process safety specialists and the pilot validation with a local management consultancy, has effectively mitigated the risk of systematic misinterpretation.

Future research could extend this work by complementing perception-based surveys with qualitative interviews or ethnographic observations to capture deeper insights into daily practices and decision-making contexts. Additionally, longitudinal studies tracking cultural evolution over time following targeted leadership interventions would provide valuable evidence on the sustainability of cultural maturity improvements. Exploring correlations between process safety culture maturity and actual incident or near-miss rates across similar regional operations could also offer more tangible validation of the cultural assessment's predictive utility.

Table 5—Appendix A, Table 14 Overall Score Table - Factor Score (Min 1 – Max 100)

Id	Questions	Factors	Factor Score	Module	Module Score	Maturity Level	Final Rank	Culture Level
1	B1	B1-I am aware of the " Core Values" of my Organization.	77.33	B	70.64	Calculative	10	74.55 Calculative
2	B2	B2-Is there a system in place that ensures a review of process safety procedures before starting any working activity involving equipment in process or hazardous area?	74.71					
3	B3	B3-Is there a system in place that ensures correct analysis and approval should a change occur during the working activity?	76.75					
4	B4	B4-In my Organization, process safety/ safety of operations is more important than Production to meet schedule	53.78					
5	C1	C1-My Organization's Senior Management is implementing visible and active leadership.	76.60	C	74.08	Calculative	6	
6	C2	C2-I clearly understand mine and my Manager/ Supervisor accountabilities and how they are enforced.	79.37					
7	C3	C3-Any identified process safety problem is immediately reported to my Manager/ Supervisor and properly analyzed and solved.	66.28					
8	D1	D1-I can report unsafe acts or conditions without fear of being punished	79.22	D	73.53	Calculative	8	
9	D2	D2-Workers are given the possibility to discuss the criticality of all process safety activities and risks before execution.	77.91					
10	D3	D3-I always challenge bad ideas when they are proposed by my colleagues during pre-job meeting	71.52					
11	D4	D4-I am aware that my manager/ supervisor uses process safety metrics to improve my performance review	65.48					
12	E1	E1-I can speak to my peers and Manager/ supervisor about my safety concerns because this is considered positive behavior	80.52	E	79.29	Proactive	1	
13	E2	E2-My organization established systems to communicate process safety information in such a way that I have easy access to expertise in all working conditions	78.05					
14	F1	F1-During hazard/ risk analyses all credible major events are duly analyzed even if they are considered not credible events	74.87	F	74.82	Calculative	5	
15	F2	F2-Process safety is my responsibility	71.10					
16	F3	F3-I saw my manager/ supervisor frequently monitoring safety rules compliance	78.49					
17	G1	G1-The recommendations issued in the hazard/ risk assessments or Job Safety analysis are correctly identified and implemented on-site	74.71	G	78.42	Proactive	2	
18	G2	G2-Process safety risks and related measures are communicated to contractors and sub-contractors	82.12					
19	H1	H1-I am empowered to propose new safety initiatives, improvements, or ideas	74.72	H	73.63	Calculative	7	
20	H2	H2-I am empowered to take corrective action as soon as possible (e.g.: shutting down equipment if necessary)	72.55					
21	I1	I1-I am sure to rely on the proper expertise during investigations of process safety incidents	77.47	I	78.13	Proactive	3	
22	I2	I2-I am sure to rely on the proper expertise when process safety issues and questions are raised before issuing work permits (e.g.: equipment isolation and opening)	78.78					
23	L1	L1: The operators always follow the proper procedures (also when the supervisor is not around)	74.73	L	70.74	Calculative	9	
24	L2	L2: I have never been asked to operate equipment and process that are unsafe	66.75					
25	M1	M1-My Organization implements a process to learn from mistakes should a process safety issue occur	76.47	M	77.19	Proactive	4	
26	M2	M2-My Organization has set up a strategy to improve the process safety culture at all levels	77.91					

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